



# Yifan Zhang

Assistant Professor, San Diego State University

Website: [1phan.com](https://1phan.com)

Location: San Diego, CA    **Cybersecurity Center @ SDSU**

*I am recruiting motivated PhD students in systems security and privacy.*

## Research Directions

My research focuses on system security and privacy, with an emphasis on identifying new attack vectors, building automated binary-analysis tools, and conducting large-scale real world measurement. Current on going projects include:

- **Software Supply Chain Security & Privacy**
- **Mobile & IoT Systems Security**
- **Network Security & Privacy**
- **Cybercrime**

I emphasize **analysis methods for systems security**; application scenarios are not restricted.

## How We Work

We find new types of attacks and forms of exploitation in current systems, then combine reverse engineering with static & dynamic program analysis, leveraging formal verification and AI where necessary to detect, measure, and mitigate novel security & privacy vulnerabilities.

## What I Look For in a PhD Student

- **Deep curiosity** about real systems and willingness to dissect complex software stacks.
- **Strong systems skills**: confident with Linux, scripting, and debugging; comfortable reading specs and source.
- **Solid coding ability**; able to build research prototypes and measurement pipelines.
- **Team fit**: collaborative, respectful, and proactive; excited to publish in top security venues.

- **Domain knowledge/interest**: experience/interest in: reverse engineering, static and dynamic analysis (e.g., program slicing, taint flow analysis, fuzzing, traffic analysis), formal methods, coding in ns-3 (network simulator 3).

## About SDSU

San Diego State University (SDSU) is a major public research university in Southern California and an **R1 university** (Very High Research Activity).

## PhD Program & Admission

Students join my lab through the [SDSU-UCI Joint PhD Program in Computational Science](#). Applicants should meet the program requirements, including:

- **Degree**: Bachelor's degree in a STEM field.
- **GPA**: Minimum 3.0 undergraduate GPA and 3.5 GPA in any prior graduate coursework.

## Interested in Joining?

If you are excited about **systems security and privacy**, and believe you may be a good fit, please prepare:

- A brief statement describing your research interests and any relevant past projects.
- A CV or résumé with links to code or artifacts.
- A link to any open-source contributions or technical write-ups you are proud of.

## Email subject template:

[SDSU Security PhD] Your Name --- [Brief Research Interest]